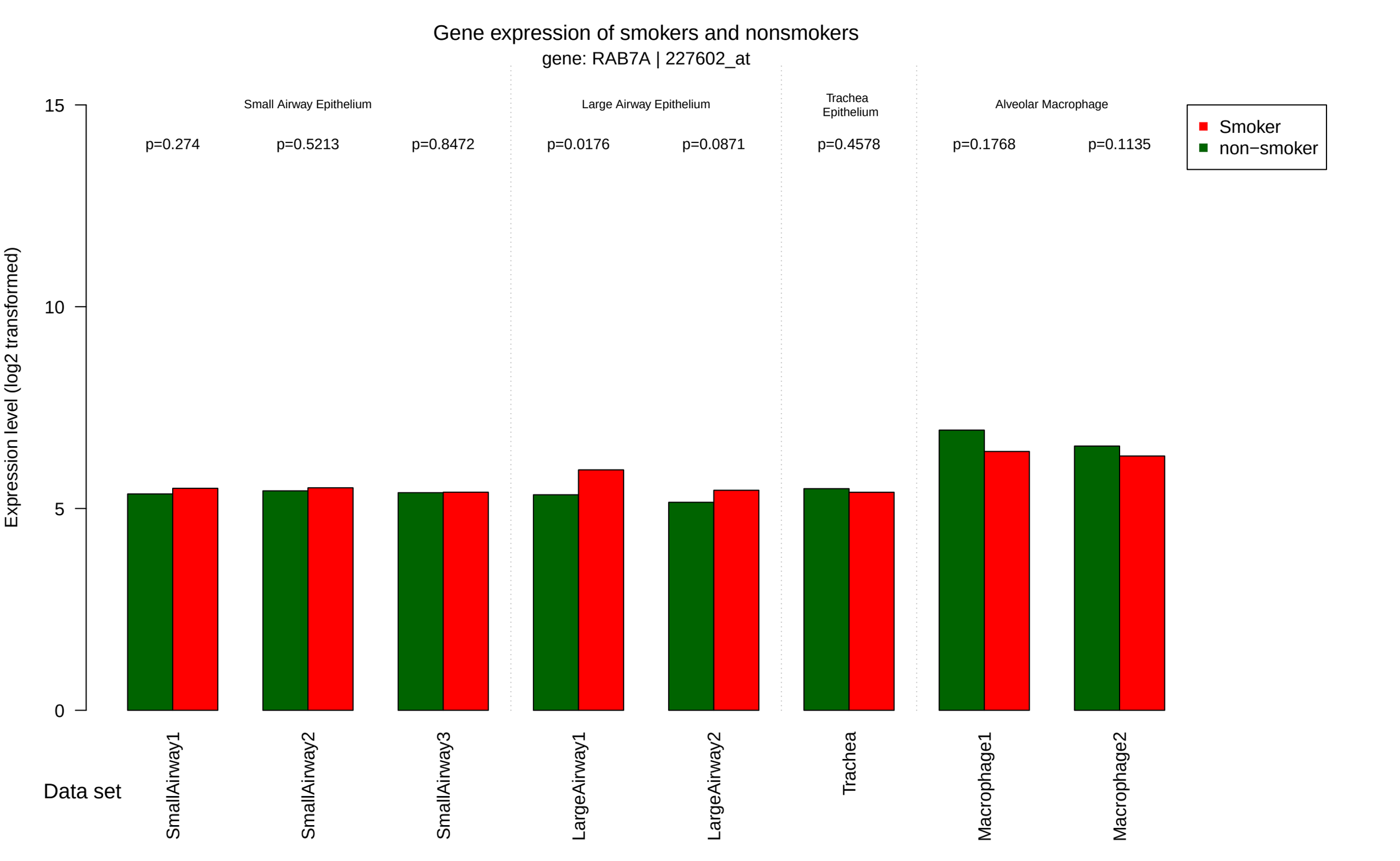
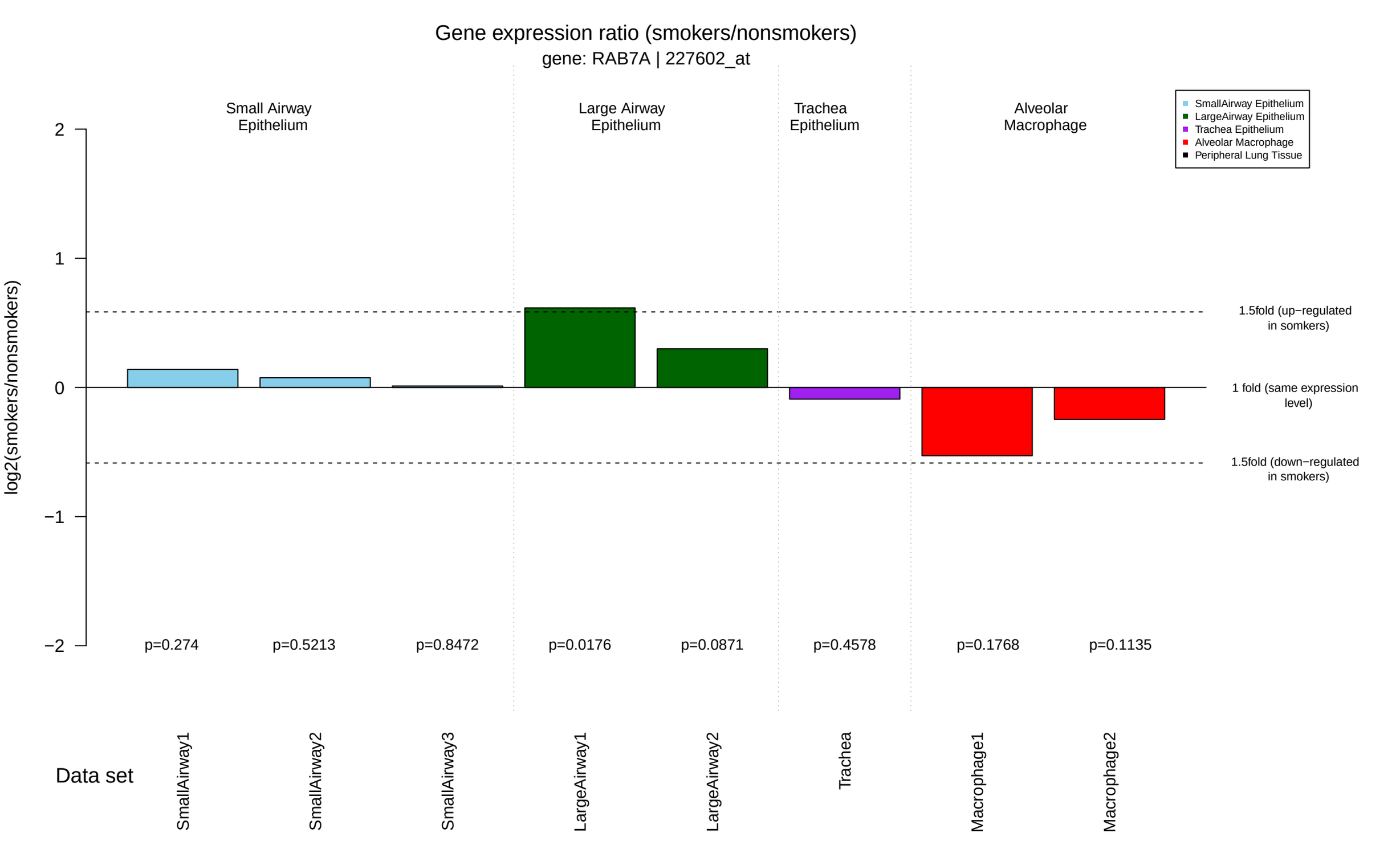
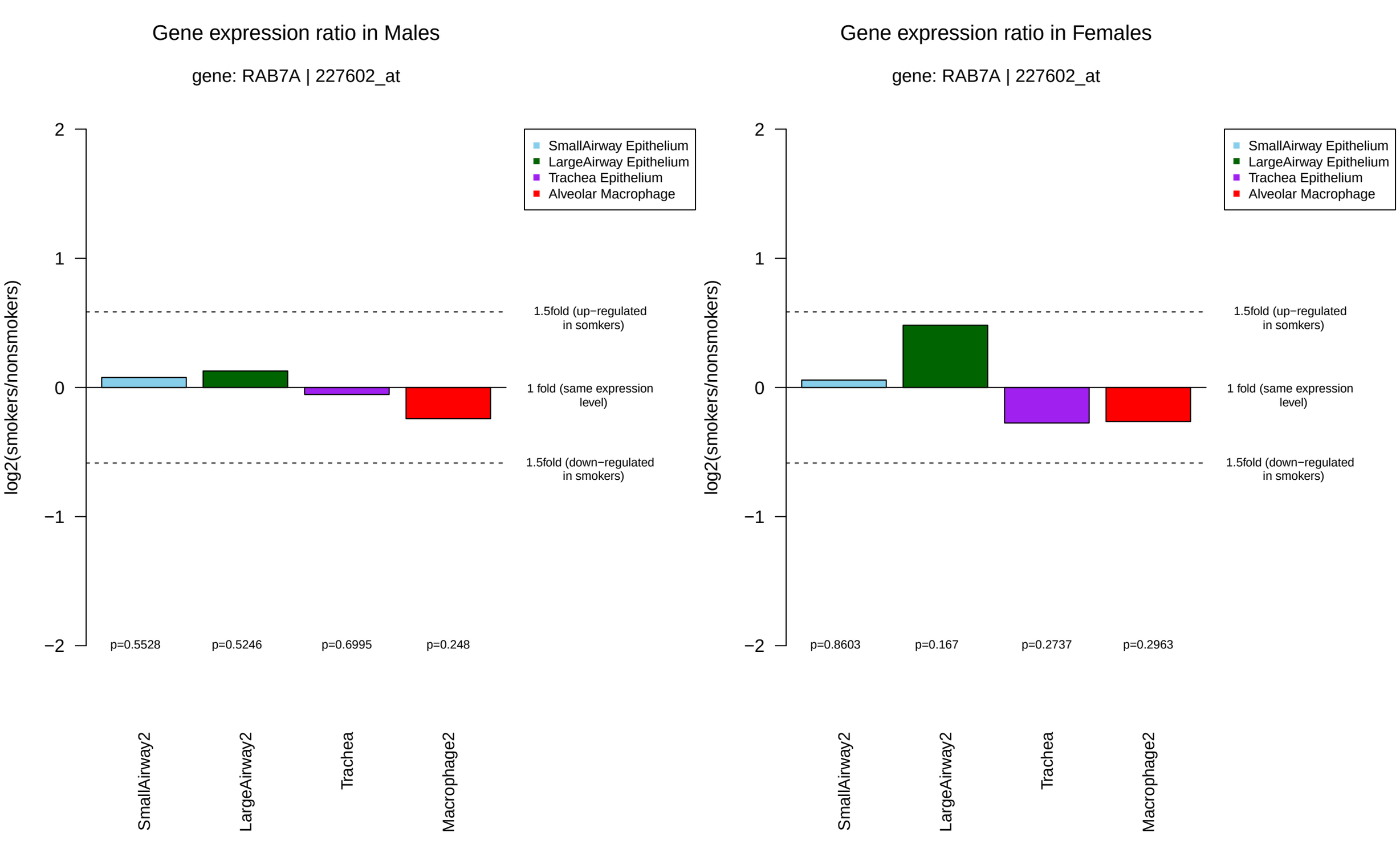
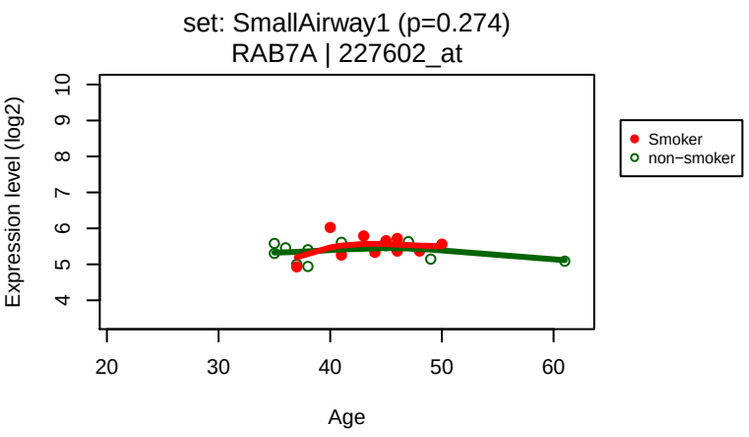
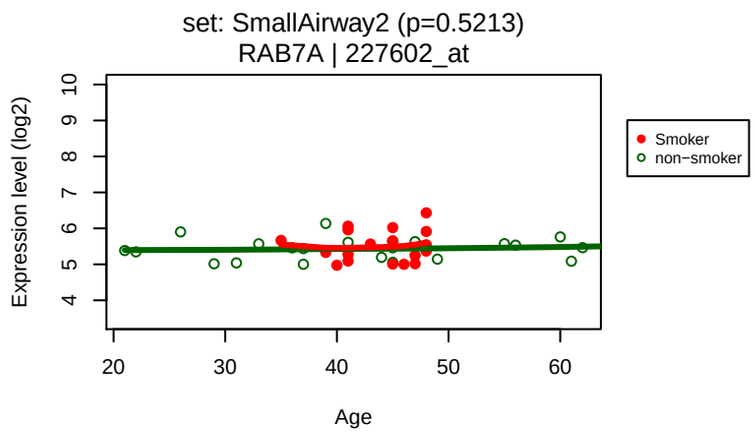
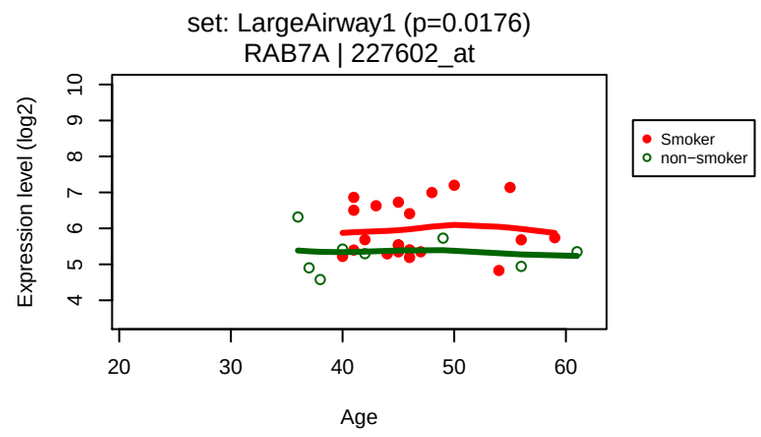

Gene expression of smokers and nonsmokers
gene: RAB7A | 227602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.274
p=0.5213
p=0.8472
p=0.0176
p=0.0871
p=0.4578
p=0.1768
p=0.1135
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

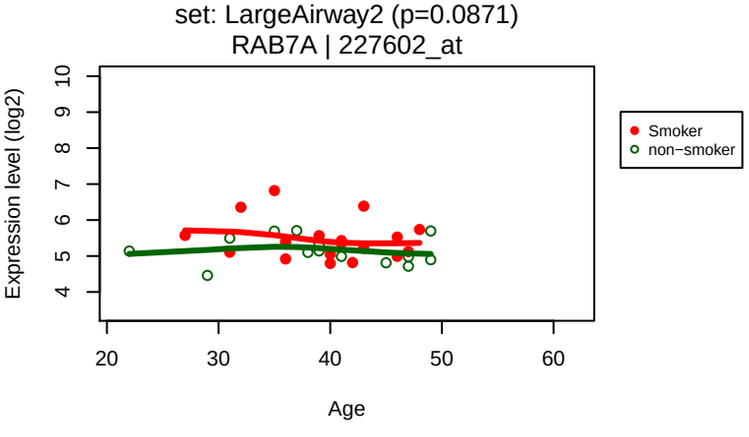

Gene expression ratio (smokers/nonsmokers)
gene: RAB7A | 227602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.274
p=0.5213
p=0.8472
p=0.0176
p=0.0871
p=0.4578
p=0.1768
p=0.1135
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

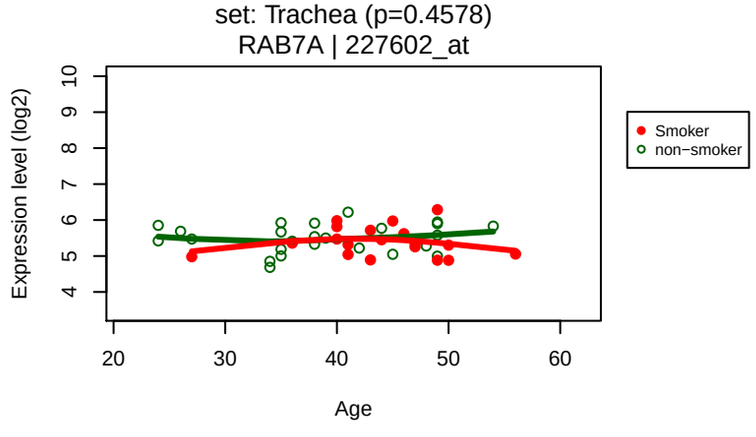

Gene expression ratio in Males
gene: RAB7A | 227602_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5528
p=0.5246
p=0.6995
p=0.248
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RAB7A | 227602_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8603
p=0.167
p=0.2737
p=0.2963
SmallAirway2
LargeAirway2
Trachea
Macrophage2

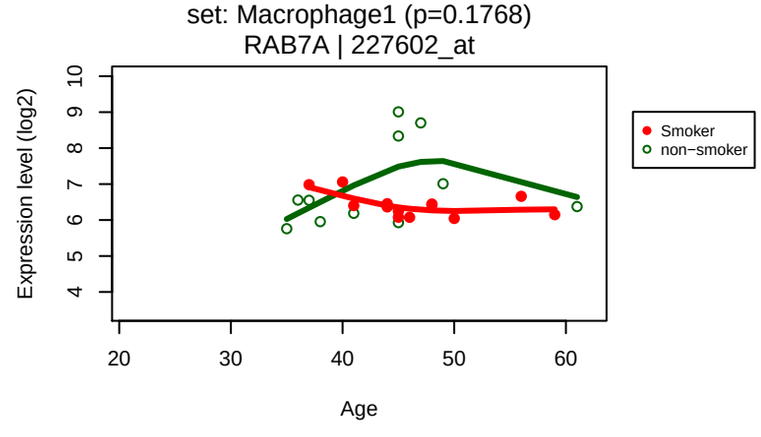

set: SmallAirway1 (p=0.274)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker

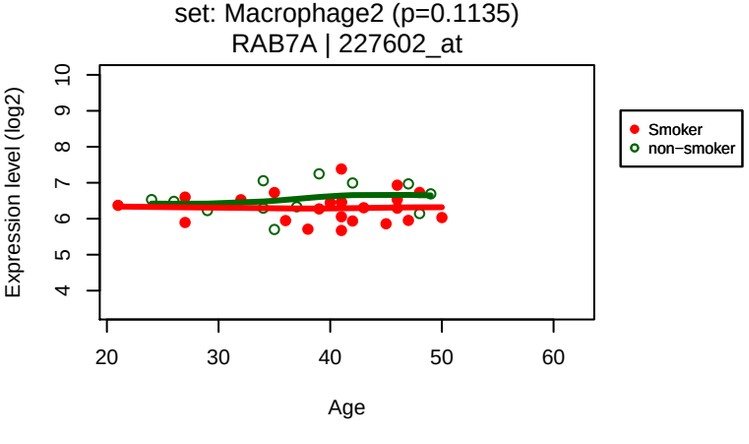

set: SmallAirway2 (p=0.5213)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker

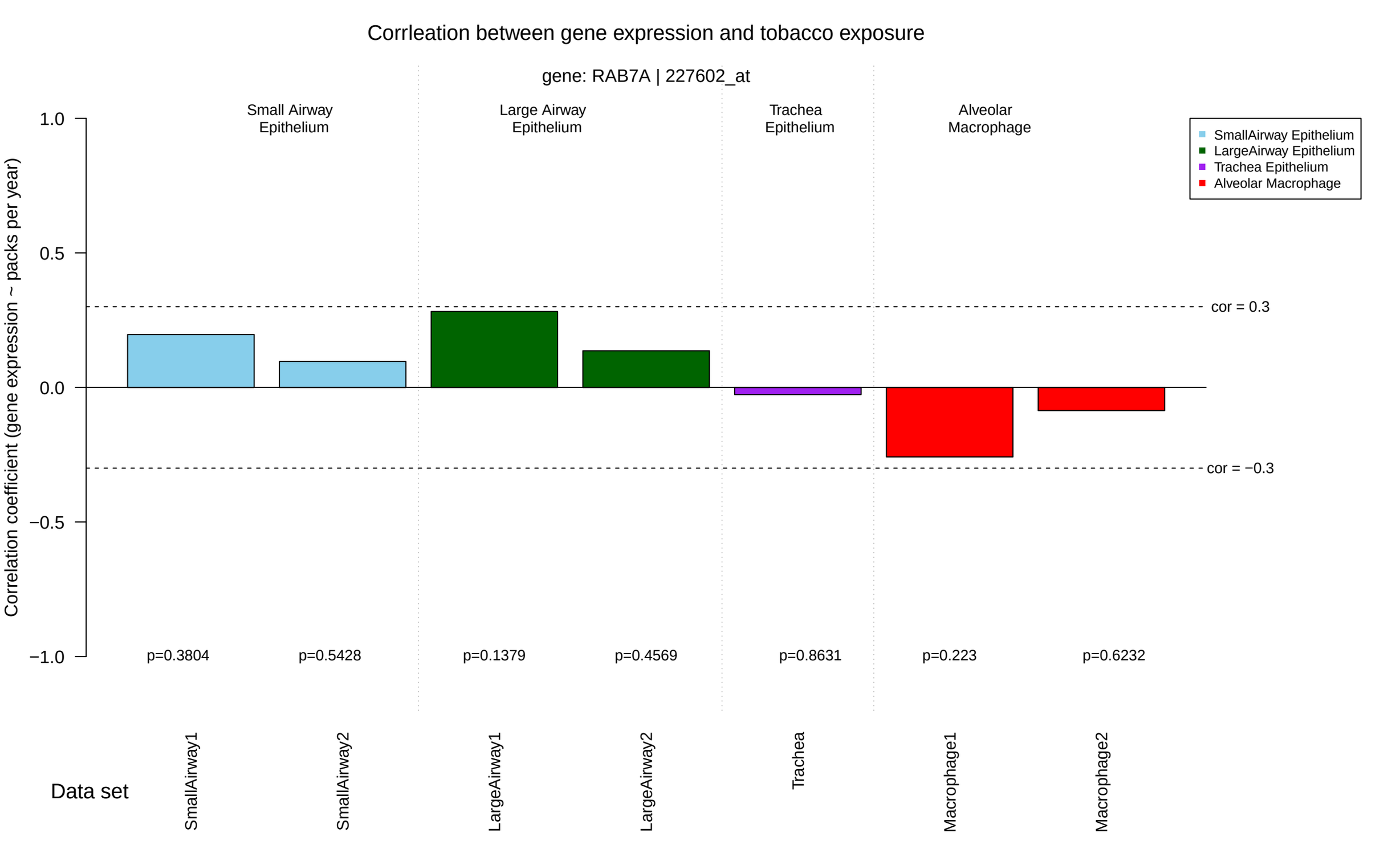

set: LargeAirway1 (p=0.0176)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0871)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4578)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1768)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1135)
RAB7A | 227602_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RAB7A | 227602_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3804
p=0.5428
p=0.1379
p=0.4569
p=0.8631
p=0.223
p=0.6232
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2